

RaceDay

API Endpoint Plan — Section B

1. Authentication

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new user account (Organiser or Participant). Validates email uniqueness and hashes the password before storing.	None (Public)	{ firstName, lastName, email, password, role }	201 Created — user object. 400 Bad Request — validation errors. 409 Conflict — email exists.
POST	/api/auth/login	Authenticates a user and returns a JWT token for use in subsequent protected requests.	None (Public)	{ email, password }	200 OK — { token, userId, role, firstName }. 401 Unauthorised — invalid credentials.
GET	/api/auth/profile	Returns the authenticated user's profile details.	Any (Logged In)	None	200 OK — user profile object. 401 Unauthorised — missing or invalid token.
PUT	/api/auth/profile	Updates the authenticated user's profile information such as name or password.	Any (Logged In)	{ firstName, lastName, password }	200 OK — updated profile. 400 Bad Request — validation errors.

2. User Profiles

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users	Returns a list of all registered users for administrative overview.	Organiser	None	200 OK — array of user objects. 403 Forbidden — insufficient role.
GET	/api/users/{id}	Returns the profile of a specific user by their ID.	Organiser	None	200 OK — user object. 404 Not Found — user does not exist.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
DELETE	/api/users/{id}	Deletes a specific user account from the system.	Organiser	None	200 OK — success message. 404 Not Found — user not found.

3. Events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Returns a list of all upcoming and past events. Participants can browse all available events.	None (Public)	None	200 OK — array of event objects (name, date, location, category, distance).
GET	/api/events/{id}	Returns full details of a specific event including route and weather information.	None (Public)	None	200 OK — event detail object. 404 Not Found — event not found.
POST	/api/events	Creates a new event. Only Event Organisers may create events.	Organiser	{ name, description, date, location, categoryId, distance, routeInfo }	201 Created — new event object. 400 Bad Request — invalid fields. 403 Forbidden — not an Organiser.
PUT	/api/events/{id}	Updates an existing event's details. Only the creating Organiser may update.	Organiser	{ name, description, date, location, categoryId, distance, routeInfo }	200 OK — updated event. 403 Forbidden — not owner. 404 Not Found.
DELETE	/api/events/{id}	Deletes an event and all associated enrolments. Only the creating Organiser may delete.	Organiser	None	200 OK — success message. 403 Forbidden — not owner. 404 Not Found.

4. Categories

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/categories	Returns all event categories (e.g. Road Running, Cycling, Walking).	None (Public)	None	200 OK — array of { id, name, description }.
GET	/api/categories/{id}	Returns the details of a specific category by ID.	None (Public)	None	200 OK — category object. 404 Not Found.
POST	/api/categories	Creates a new event category.	Organiser	{ name, description }	201 Created — new category. 400 Bad Request.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
					409 Conflict — name exists.
PUT	/api/categories/{id}	Updates an existing category's name or description.	Organiser	{ name, description }	200 OK — updated category. 404 Not Found.
DELETE	/api/categories/{id}	Deletes a category. Fails if events are still linked to it.	Organiser	None	200 OK — success. 409 Conflict — category in use.

5. Event Enrolments

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/enrolments	Returns all enrolments across all events for Organiser management.	Organiser	None	200 OK — array of enrolment objects with participant and event details.
GET	/api/enrolments/my	Returns all events the authenticated Participant is enrolled in.	Participant	None	200 OK — enrolment array. 401 Unauthorised — not logged in.
GET	/api/enrolments/event/{eventId}	Returns all participants enrolled in a specific event.	Organiser	None	200 OK — array of participant enrolment records. 404 Not Found.
POST	/api/enrolments	Enrolls the authenticated Participant in a specified event. Prevents duplicate enrolments.	Participant	{ eventId }	201 Created — enrolment confirmation. 409 Conflict — already enrolled. 404 Not Found.
DELETE	/api/enrolments/{id}	Cancels a specific enrolment. Participants may only cancel their own.	Participant	None	200 OK — success. 403 Forbidden — not owner. 404 Not Found.

6. Results

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/results	Returns all recorded race results across all events.	None (Public)	None	200 OK — array of result objects (participant, event, finish time, position).

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/results/event/{eventId}	Returns results for a specific event ordered by finish time (leaderboard).	None (Public)	None	200 OK — ordered result array. 404 Not Found.
GET	/api/results/my	Returns the personal performance history of the authenticated Participant.	Participant	None	200 OK — result array for logged-in user. 401 Unauthorised.
POST	/api/results	Records a result for a participant in a specific event.	Organiser	{ eventId, userId, finishTime, position }	201 Created — result object. 400 Bad Request. 404 Not Found. 409 Conflict — result already exists.
PUT	/api/results/{id}	Updates an existing result (e.g. to correct a finish time or position).	Organiser	{ finishTime, position }	200 OK — updated result. 404 Not Found.
DELETE	/api/results/{id}	Deletes a specific result record.	Organiser	None	200 OK — success. 404 Not Found.

7. Weather & Route Information

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{id}/weather	Returns live or forecast weather data for the event location via an external weather API.	None (Public)	None	200 OK — { temperature, conditions, humidity, windSpeed, forecast }. 404 Not Found. 503 Service Unavailable.
GET	/api/events/{id}/route	Returns route information for a specific event including distance, elevation, and map data.	None (Public)	None	200 OK — { distance, elevationGain, mapUrl, waypoints[] }. 404 Not Found.

8. Summary

Area	Endpoints	Public	Participant	Organiser
Authentication	4	2	1	1
User Profiles	3	0	0	3
Events	5	2	0	3

Area	Endpoints	Public	Participant	Organiser
Categories	5	2	0	3
Enrolments	5	0	3	2
Results	6	2	1	3
Weather & Route	2	2	0	0
Total	30	10	5	15